

Team 7 - E-Waste Battery Extraction

CV-23 Part A: Downstream Detection Consumer & ROS 2 Interface Integration

Implementation and Interface Validation

ROS 2 Jazzy, Gazebo, custom detection messages and downstream pose integration

Ubuntu 24.04 on WSL2

Downstream consumer implementation report

September 2026

Contents

1	Purpose of CV-23 Part A	2
2	System Context	2
3	Verified Input Interface	2
3.1	PhoneDetection	2
3.2	Frame-Level Collection	3
4	Consumer Design	3
4.1	Planning-Friendly Output	4
5	ROS 2 Packaging and Build	4
6	Live Integration Test	4
7	Invalid-Detection Rejection	6
8	Integration Issue Encountered	7
9	Part A Outcome	7
A	Core Reproduction Commands	7
A.1	Start the consumer	7
A.2	Inspect the interface	7
A.3	Inspect the live chain	7
B	Part A Artefacts	8

1 Purpose of CV-23 Part A

CV-23 is the downstream detection consumer integration task for the Team 7 perception pipeline. The task is not to create another detector. Its purpose is to prove that the CV output can be consumed by a separate ROS 2 component and converted into information that a planning or robotics component can use.

Part A focused on the perception-to-consumer boundary. The target path was:

`/phone_detections → detection_consumer → /cv_target_pose`

The implementation used the existing world-coordinate fields produced by the earlier localisation work. This was preferable to reimplementing projection inside CV-23 because the consumer should depend on the agreed detection interface rather than duplicate perception geometry.

2 System Context

The work was completed in the Team 7 WSL2 environment using ROS 2 Jazzy and Gazebo. The active perception workspace was:

`/mnt/e/team7_project_bundle/team7_perception_ws`

The relevant upstream components were already available:

- the overhead Gazebo camera and camera information;
- the YOLO11n-seg ONNX detector;
- the custom `team7_msgs/msg/PhoneDetectionArray` output;
- the CV-17 pixel-to-world localisation fields `world_x`, `world_y`, `world_z`, and `world_valid`;
- the permanent `world <- top_camera_optical_frame` transform used by localisation.

The CV-23 consumer was therefore designed as a separate downstream node rather than as another block inside the detector callback.

3 Verified Input Interface

3.1 PhoneDetection

The installed message interface was inspected before implementation. The important fields were:

Field	Purpose
<code>class_id</code>	detected class label
<code>score</code>	confidence in the range 0–1
<code>bbox_*</code>	axis-aligned image-space bounding box
<code>mar_center_x/y</code>	oriented-box centroid in image pixels
<code>world_x/y/z</code>	projected position in Gazebo world coordinates
<code>world_valid</code>	indicates whether the world projection is usable

The message still contained a stale comment referring to a 320x240 source image even though the active overhead camera is 1280x960. This did not affect CV-23 because the consumer uses the explicit world-coordinate fields rather than assuming a pixel resolution.

```
ad@DESKTOP-0T45K60:/mnt/e/team7_project_bundle/team7_perception_ws$ ros2 interface show team7_msgs/msg/PhoneDetection
std_msgs/Header header
  builtin_interfaces/Time stamp
    int32 sec
    uint32 nanosec
  string frame_id
string class_id          # from the model's class map, e.g. "battery"
float64 score             # 0..1

# pixel-space, in the ORIGINAL /top_camera/image resolution (320x240)
int32 bbox_x1
int32 bbox_y1
int32 bbox_x2
int32 bbox_y2

# oriented box from the segmentation mask (cv2.minAreaRect), pixel-space
float64 mar_center_x
float64 mar_center_y
float64 mar_width
float64 mar_height
float64 angle_deg

# projected world-frame position (ground-plane assumption, see geometry.py)
float64 world_x
float64 world_y
float64 world_z
bool world_valid         # false if camera_info not yet received, etc.
```

Figure 1: Installed PhoneDetection interface inspected before writing the consumer.

3.2 Frame-Level Collection

The array message contains one header and a list of detections. This allows one consumer callback to inspect every candidate from the same processed image and also supports an empty collection when no object is detected.

```
ad@DESKTOP-0T45K60:/mnt/e/team7_project_bundle/team7_perception_ws$ ros2 interface show team7_msgs/msg/PhoneDetectionArray
std_msgs/Header header
  builtin_interfaces/Time stamp
    int32 sec
    uint32 nanosec
  string frame_id
PhoneDetection[] detections
std_msgs/Header header
  builtin_interfaces/Time stamp
    int32 sec
    uint32 nanosec
  string frame_id
  string class_id          #
float64 score              #
int32 bbox_x1
int32 bbox_y1
int32 bbox_x2
int32 bbox_y2
float64 mar_center_x
float64 mar_center_y
float64 mar_width
float64 mar_height
float64 angle_deg
float64 world_x
float64 world_y
float64 world_z
bool world_valid          #
```

Figure 2: Installed PhoneDetectionArray interface used as the CV-23 input contract.

4 Consumer Design

The new node was created at:

```
team7_perception_ws/src/team7_perception/
team7_perception/detection_consumer_node.py
```

The node exposes four parameters:

```
input_topic   = /phone_detections
output_topic  = /cv_target_pose
target_class  = battery
min_score     = 0.40
```

Each callback filters the input array using three conditions:

```
valid_detections = [
    detection
    for detection in msg.detections
    if detection.class_id == self.target_class
    and detection.score >= self.min_score
    and detection.world_valid
]
```

If more than one valid battery is present, the highest-confidence candidate is selected. If no candidate passes all conditions, the callback returns without publishing a target.

4.1 Planning-Friendly Output

The accepted detection is converted into `geometry_msgs/msg/PoseStamped` and published on `/cv_target_pose`. The position is copied directly from the validated world-coordinate fields:

```
pose.header.stamp = msg.header.stamp
pose.header.frame_id = 'world'
pose.pose.position.x = target.world_x
pose.pose.position.y = target.world_y
pose.pose.position.z = target.world_z
pose.pose.orientation.w = 1.0
```

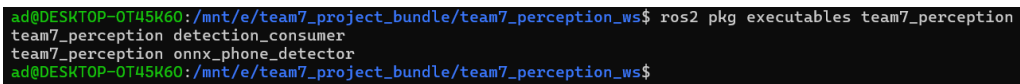
The source timestamp is preserved, while the output frame is explicitly set to `world`. Identity orientation is used only as a neutral placeholder because the CV interface currently supplies position rather than a grasp orientation.

5 ROS 2 Packaging and Build

The consumer was registered as a package executable:

```
"detection_consumer = team7_perception.detection_consumer_node:main"
```

The package manifest was also updated to declare `geometry_msgs`. The new source passed `python3 -m py_compile`, the package rebuilt successfully, and ROS 2 reported both the original detector and the new consumer executable.



```
ad@DESKTOP-0T45K60:/mnt/e/team7_project_bundle/team7_perception_ws$ ros2 pkg executables team7_perception
team7_perception detection_consumer
team7_perception onnx_phone_detector
ad@DESKTOP-0T45K60:/mnt/e/team7_project_bundle/team7_perception_ws$
```

Figure 3: ROS 2 package executable list after adding the CV-23 consumer.

6 Live Integration Test

The live stack used the following processes:

1. Gazebo simulation and camera bridge;
2. localisation TF;
3. ONNX detector;
4. CV-23 detection consumer;
5. a separate terminal for topic inspection.

The consumer started successfully and repeatedly accepted a live battery-class detection with confidence approximately 0.907. The callback reported both the image centroid and the world-coordinate target.

```
ad@DESKTOP-0T4SK60:/mnt/e/team7_project_bundle/team7_perception_ws$ ros2 run team7_perception detection_consumer
[INFO] [1789887286.827973496]: [detection_consumer]: CV-23 consumer started: /phone_detections -> /cv_target_pose
[INFO] [1789887286.86868186]: [detection_consumer]: Target accepted: class=battery, min_score=0.48
[INFO] [1789887286.424776751]: [detection_consumer]: Target accepted: class=battery, score=0.907, pixel=(1078.5, 464.5), world=(0.0158, 0.1531, 0.7700)
[INFO] [1789887287.383387131]: [detection_consumer]: Target accepted: class=battery, score=0.907, pixel=(1078.5, 464.5), world=(0.0158, 0.1531, 0.7700)
[INFO] [1789887288.512921607]: [detection_consumer]: Target accepted: class=battery, score=0.907, pixel=(1078.5, 464.5), world=(0.0158, 0.1531, 0.7700)
[INFO] [1789887289.545409519]: [detection_consumer]: Target accepted: class=battery, score=0.907, pixel=(1078.5, 464.5), world=(0.0158, 0.1531, 0.7700)
[INFO] [1789887210.487821331]: [detection_consumer]: Target accepted: class=battery, score=0.907, pixel=(1078.5, 464.5), world=(0.0158, 0.1531, 0.7700)
[INFO] [1789887211.349084230]: [detection_consumer]: Target accepted: class=battery, score=0.907, pixel=(1078.5, 464.5), world=(0.0158, 0.1531, 0.7700)
[INFO] [1789887212.436982126]: [detection_consumer]: Target accepted: class=battery, score=0.907, pixel=(1078.5, 464.5), world=(0.0158, 0.1531, 0.7700)
[INFO] [1789887213.263720506]: [detection_consumer]: Target accepted: class=battery, score=0.907, pixel=(1078.5, 464.5), world=(0.0158, 0.1531, 0.7700)
[INFO] [1789887214.278974610]: [detection_consumer]: Target accepted: class=battery, score=0.907, pixel=(1078.5, 464.5), world=(0.0158, 0.1531, 0.7700)
[INFO] [1789887215.364518517]: [detection_consumer]: Target accepted: class=battery, score=0.907, pixel=(1078.5, 464.5), world=(0.0158, 0.1531, 0.7700)
[INFO] [1789887216.398756221]: [detection_consumer]: Target accepted: class=battery, score=0.907, pixel=(1078.5, 464.5), world=(0.0158, 0.1531, 0.7700)
[INFO] [1789887217.699286515]: [detection_consumer]: Target accepted: class=battery, score=0.907, pixel=(1078.5, 464.5), world=(0.0158, 0.1531, 0.7700)
[INFO] [1789887218.814963995]: [detection_consumer]: Target accepted: class=battery, score=0.907, pixel=(1078.5, 464.5), world=(0.0158, 0.1531, 0.7700)
[INFO] [1789887220.126134983]: [detection_consumer]: Target accepted: class=battery, score=0.907, pixel=(1078.5, 464.5), world=(0.0158, 0.1531, 0.7700)
[INFO] [1789887221.453642730]: [detection_consumer]: Target accepted: class=battery, score=0.907, pixel=(1078.5, 464.5), world=(0.0158, 0.1531, 0.7700)
[INFO] [1789887222.628290369]: [detection_consumer]: Target accepted: class=battery, score=0.907, pixel=(1078.5, 464.5), world=(0.0158, 0.1531, 0.7700)
[INFO] [1789887223.720212240]: [detection_consumer]: Target accepted: class=battery, score=0.907, pixel=(1078.5, 464.5), world=(0.0158, 0.1531, 0.7700)
[INFO] [1789887224.886004357]: [detection_consumer]: Target accepted: class=battery, score=0.907, pixel=(1078.5, 464.5), world=(0.0158, 0.1531, 0.7700)
[INFO] [1789887226.181676544]: [detection_consumer]: Target accepted: class=battery, score=0.907, pixel=(1078.5, 464.5), world=(0.0158, 0.1531, 0.7700)
[INFO] [1789887227.435449445]: [detection_consumer]: Target accepted: class=battery, score=0.907, pixel=(1078.5, 464.5), world=(0.0158, 0.1531, 0.7700)
[INFO] [1789887228.394136841]: [detection_consumer]: Target accepted: class=battery, score=0.907, pixel=(1078.5, 464.5), world=(0.0158, 0.1531, 0.7700)
[INFO] [1789887229.395160134]: [detection_consumer]: Target accepted: class=battery, score=0.907, pixel=(1078.5, 464.5), world=(0.0158, 0.1531, 0.7700)
[INFO] [1789887230.578781471]: [detection_consumer]: Target accepted: class=battery, score=0.907, pixel=(1078.5, 464.5), world=(0.0158, 0.1531, 0.7700)
[INFO] [1789887231.580531398]: [detection_consumer]: Target accepted: class=battery, score=0.907, pixel=(1078.5, 464.5), world=(0.0158, 0.1531, 0.7700)
[INFO] [1789887232.689649165]: [detection_consumer]: Target accepted: class=battery, score=0.907, pixel=(1078.5, 464.5), world=(0.0158, 0.1531, 0.7700)
[INFO] [1789887233.697935951]: [detection_consumer]: Target accepted: class=battery, score=0.907, pixel=(1078.5, 464.5), world=(0.0158, 0.1531, 0.7700)
[INFO] [1789887234.576495650]: [detection_consumer]: Target accepted: class=battery, score=0.907, pixel=(1078.5, 464.5), world=(0.0158, 0.1531, 0.7700)
[INFO] [1789887235.908544834]: [detection_consumer]: Target accepted: class=battery, score=0.907, pixel=(1078.5, 464.5), world=(0.0158, 0.1531, 0.7700)
[INFO] [1789887236.808750213]: [detection_consumer]: Target accepted: class=battery, score=0.907, pixel=(1078.5, 464.5), world=(0.0158, 0.1531, 0.7700)
```

Figure 4: Live CV-23 consumer receiving /phone_detections and publishing a downstream target.

The source message contained:

Quantity	Live value
Class	battery
Score	0.9074
MAR centroid	(1078.5, 464.5) px
World X	0.015797 m
World Y	0.153102 m
World Z	0.770000 m
World valid	true

```

ad@DESKTOP-0T45K60:/mnt/e/team7_project_bundle/team7_perception_ws$ ros2 topic echo /phone_detections --once
header:
  stamp:
    sec: 347
    nanosec: 500000000
  frame_id: overhead_camera/mount/overhead_camera
detections:
- header:
  stamp:
    sec: 347
    nanosec: 500000000
  frame_id: overhead_camera/mount/overhead_camera
  class_id: battery
  score: 0.9073662757873535
  bbox_x1: 880
  bbox_y1: 19
  bbox_x2: 1278
  bbox_y2: 911
  mar_center_x: 1078.5
  mar_center_y: 464.49993896484375
  mar_width: 396.99993896484375
  mar_height: 890.9998779296875
  angle_deg: 90.0
  world_x: 0.01579691140254362
  world_y: 0.15310204041751047
  world_z: 0.77
  world_valid: true
---
```

Figure 5: Live source detection message used by the downstream consumer.

The output topic then carried the same world position in a PoseStamped with frame_id: world.

```

ad@DESKTOP-0T45K60:/mnt/e/team7_project_bundle/team7_perception_ws$ ros2 topic echo /cv_target_pose --once
header:
  stamp:
    sec: 352
    nanosec: 0
  frame_id: world
pose:
  position:
    x: 0.01579691140254362
    y: 0.15310204041751047
    z: 0.77
  orientation:
    x: 0.0
    y: 0.0
    z: 0.0
    w: 1.0
---
```

Figure 6: Downstream /cv_target_pose output. The world coordinates match the accepted input detection.

This establishes the key Part A result: the agreed CV output is consumable by an independent ROS 2 node and can be translated into a standard planning-oriented pose message.

7 Invalid-Detection Rejection

A separate test topic was used so the live pipeline did not need to be modified. A high-confidence battery detection was published with valid-looking world coordinates but world_valid=false. The consumer rejected the message and logged:

No valid downstream target in this frame.

```

ad@DESKTOP-0T45K60:~$ ros2 run team7_perception detection_consumer --ros-args \
-p input_topic:=/cv23_test_detections \
-p output_topic:=/cv23_test_target
[INFO] [1789811670.937749451] [detection_consumer]: CV-23 consumer started: /cv23_test_detections -> /cv23_test_target
[INFO] [1789811670.999549986] [detection_consumer]: Target class="battery", min_score=0.40
[INFO] [1789811698.002776368] [detection_consumer]: No valid downstream target in this frame.

```

Figure 7: Negative test: the consumer rejects a high-confidence battery when world_valid=false.

This test is important because downstream planning must not treat numeric coordinates as valid simply because they are present in a message.

8 Integration Issue Encountered

During bring-up, the E-drive copy of the workspace did not contain the CV-17 `localisation_tf.launch.py` source file even though the completed WSL-home workspace did. The tested launch file was recovered from the earlier workspace, copied into the current package, and added to `setup.py` so it was installed correctly by `colcon`. This restored the required world-to-camera transform without recreating or changing the validated localisation geometry.

9 Part A Outcome

Part A produced a functioning downstream ROS 2 interface rather than a terminal-only demonstration. The consumer:

- subscribes to the real CV detection collection;
- filters by class, confidence and localisation validity;
- selects a deterministic target candidate;
- publishes the accepted world position as a standard `PoseStamped`;
- preserves the source timestamp and explicitly labels the output world frame;
- rejects invalid localisation results instead of forwarding them.

The remaining question for Part B was whether this pose could be used by the existing MoveIt/Kinova pick-place component rather than only observed with `ros2 topic echo`.

A Core Reproduction Commands

A.1 Start the consumer

```
cd /mnt/e/team7_project_bundle/team7_perception_ws
source /opt/ros/jazzy/setup.bash
source install/setup.bash
ros2 run team7_perception detection_consumer
```

A.2 Inspect the interface

```
ros2 interface show team7_msgs/msg/PhoneDetection
ros2 interface show team7_msgs/msg/PhoneDetectionArray
```

A.3 Inspect the live chain

```
ros2 topic echo /phone_detections --once
ros2 topic echo /cv_target_pose --once
```

B Part A Artefacts

The final CV-23 handover bundle preserves the consumer implementation together with the two installed interface definitions. The implementation was also retained in the active perception workspace so it can be launched as a normal ROS 2 executable.